EventSync: Multi-Client Event Scheduling and Resource Management

MINI PROJECT ESE HACKATHON 2026 | Development window: 10–12 hours | 100 marks

The challenge

EventSync is an event management company that organizes conferences, workshops, exhibitions and product launches for different clients. It manages multiple venues, each with halls of different capacities and a limited pool of shared equipment and support staff. Overlapping requests, last-minute changes and manual tracking sometimes lead to double bookings or resource shortages.

Build a web-based application that accepts client event requests, checks scheduling and resource constraints, allocates suitable halls and resources, and manages booking changes reliably.

Available resources (initial dataset)

Venue	Halls (seating capacity)	Shared resources at the venue
A	A1: 60; A2: 150	2 projectors; 2 audio systems; 5 staff
B	B1: 100; B2: 200	2 projectors; 1 audio system; 4 staff
C	C1: 80; C2: 250	3 projectors; 2 audio systems; 6 staff

Required functionality and constraints

- Client and events: Record clients and event requests with date, start/end time, expected attendees, preferred venue, and quantities of projectors, audio systems and support staff. View booking status.
- Hall selection: Select the smallest-capacity hall that satisfies attendance and availability. First search the preferred venue; if it has no feasible hall/resources, search the other venues. Use a consistent tie-breaker when choices are equal.
- Time restrictions: Operating hours are 8:00 AM–10:00 PM. Each booking reserves its hall and resources from 30 minutes before event start until 30 minutes after event end. This entire interval must fall within operating hours.
- Shared resources: Concurrent bookings at the same venue must not exceed available equipment or staff. A booking is confirmed only if a hall and all requested resources are available for its full reserved interval.
- Changes: Support cancellations (release resources) and rescheduling (recheck every constraint). If rescheduling fails, keep the original confirmed booking unchanged.
- Decision visibility: Show confirmed bookings, allocated hall/resources, and an understandable reason whenever a request cannot be accommodated. Maintain records in a database.

Sample situation

Venue A receives a conference (120 guests, 10:00 AM–1:00 PM, 1 projector, 3 staff) and a workshop (50 guests, 11:00 AM–2:00 PM, 1 projector, 2 staff). Can both be confirmed? What if a third client requests a 100-person product launch at 1:15 PM? Consider hall capacity, shared resources, and setup/cleanup—not just the event timings.

Innovation challenge

Each team is encouraged to implement at least one additional, useful feature of its own choice. Explain the real-world need it addresses and demonstrate its implementation. Relevant, well-executed enhancements can distinguish stronger solutions within the existing evaluation criteria.

Technology and deliverables

Use any suitable stack: React, Angular or HTML/CSS/JavaScript; Node.js/Express, Python or Java-based backends; and MySQL, PostgreSQL, SQLite, MongoDB or equivalent. Framework selection and visual polish are not the main assessment focus. Present a working prototype, database-backed booking workflow, conflict handling, change management, and an explanation of core data structures, algorithms and design decisions. External services, payments, IoT and cloud deployment are not required.

Teams may use images, icons, and other visual assets sourced from the web to improve the interface, provided applicable usage rights and attribution requirements are respected. The focus of evaluation remains on core logic, data management, and working functionality.

Official evaluation rubric

Criterion	**Marks**
Problem Understanding and Requirements Analysis	15
Algorithm Design and Problem-Solving	20
Technical Implementation and Code Quality	30
Project Demonstration	15
Responses to Judges' Questions	20
TOTAL	100